Math708 – Foundations of Computational Mathematics I

Fall 2012

Homework 2

1. Let a partition $\triangle : a = x_0 < x_1 < \cdots < x_n = b$ of the interval $[a, b]$ as well as support ordinates $f_0, f_1, \cdots, f_n \in \mathbb{R}$ be given.
   a) Show that for each number $f'_0 \in \mathbb{R}$ there exists a unique interpolating quadratic spline s_2 that satisfies the additional condition $s'_2(x_0) = f'_0$. Provide an algorithm to calculate s_2.
   b) Find the interpolating quadratic spline s_2 with periodic boundary conditions $s'_2(x_0) = s'_2(x_n)$. Make a statement on existence and uniqueness of s_2.

2. Let a partition $\triangle : a = x_0 < x_1 < \cdots < x_n = b$ of the interval $[a, b]$ as well as a function $f \in \mathbb{C}^1[a, b]$ be given. A function $s_H \in \mathbb{C}^1[a, b]$ is called the **Hermite** cubic spline interpolating f at $\{x_i\}_{i=0}^n$ if

$$\begin{cases} s_H|_{[x_i, x_i+1]} \in \mathbb{P}_3, \quad i = 0, 1, \cdots, n-1 \\ s_H(x_i) = f(x_i), s'_H(x_i) = f'(x_i), \quad i = 0, 1, \cdots, n. \end{cases}$$

   Now suppose that f is a polynomial of degree 3. Show that the **Hermite** cubic spline s_H which interpolates f at $\{x_i\}_{i=0}^n$ is identical to f (i.e., $s_H \equiv f$).

3. (**Programming Exercise for Spline**) Write a program to interpolate the function

$$f(x) = \frac{1}{25x^2 + 1}, \quad x \in [-1, 1],$$

   using a cubic spline $s_3(x)$ at the equidistant points $x_j = -1 + 2j/n, \ \ j = 0, 1, \cdots, n$. with

   - Natural boundary conditions: $s''_3(-1) = s''_3(1) = 0$.
   - Periodic boundary conditions: $s'_3(-1) = s'_3(1)$ and $s''_3(-1) = s''_3(1)$.

   Plot the graphs and calculate $\|s_3 - f\|_\infty$ respectively for $n = 5, 10$. Also submit your codes in both paper and email.

4. For even N, let $(N + 1)$ support points $a \le x_0 < x_1 < \cdots < x_N < a + 2\pi$ and support ordinates $f_0, \cdots, f_N \in \mathbb{C}$ be given. Show that:
   a) There is exactly one trigonometric polynomial of the form

$$T(x) = A_0/2 + \sum_{j=1}^{N/2} \big(A_j \cos(jx) + B_j \sin(jx)\big)$$

   with $T(x_k) = f_k$ for $k = 0, 1, \cdots, N$.
   b) If all the support ordinates $f_0, \cdots, f_N$ are real, the the coefficients A_j, B_j are also real.

5. Let the mapping $\mathbf{D} : \mathbb{C}^N \to \mathbb{C}^N$ defined by

$$\mathbf{Dc} = (-c_{k-1} + 2c_k - c_{k+1})_{k=0}^{N-1}$$

for any vector $\mathbf{c} = (c_0, c_1, \cdots, c_{N-1})^T \in \mathbb{C}^N$ with $c_{-1} = c_{N-1}$ and $c_N = c_0$. Let the diagonal matrix $M = \mathrm{diag}(\lambda_0, \lambda_1, \cdots, \lambda_{N-1}) \in \mathbb{C}^{N\times N}$ where $\lambda_j = 4\sin^2(j\pi/N) \in \mathbb{R}$ for $j = 0, 1, \cdots, N-1$, and vectors $\mathbf{v}^{(j)} = \left(e^{i2\pi jk/N}\right)_{k=0}^{N-1}$ for $j = 0, 1, \cdots, N-1$. Then there holds

$$\mathbf{D} = \mathcal{F}^{-1} M \mathcal{F}.$$

where $\mathcal{F}$ denotes the *discrete Fourier transform.*
(*Hint*: Show that $\mathbf{Dc} = \dfrac{1}{N}\sum_{j=0}^{N-1} \lambda_j \left((\mathbf{v}^{(j)})^H \mathbf{c}\right) \mathbf{v}^{(j)}$ for any $\mathbf{c} \in \mathbb{C}^N$).

6. (**Programming Exercise for FFT**) For any function $f(x)$ defined on $[0, L]$, there exists an interpolation of f using the real trigonometric polymonials of the form

$$T(x) = A_0 + 2\sum_{k=1}^{N/2-1} \Big((A_k \cos(\frac{k2\pi x}{L}) + B_k \sin(\frac{k2\pi x}{L})\Big) + A_{N/2}\cos(\frac{N\pi x}{L})$$

with even number N and equidistant support points $x_j = jL/N$ for $j = 0, 1, \cdots, N-1$. The coefficients A_k,B_k can be determined indirectly through the fast Fourier transform (See the lecture notes). Interpolate the following two functions

- $f(x) = x^2(1-x)^2, \quad x \in [0, 1].$
- $f(x) = e^{2x}, \quad x \in [0, 2].$

with $N = 2^m$, $m = 2, 4, 6$. Plot the figures respectively and calculate the maximum of absolute occurring errors at the points (to be linearly transformed) $x_j = jL/40$ for $j = 1, 2, \cdots, 40$.